

CS & IT ENGINEERING



Computer Network

IPv4 Address

Lecture No. - 06



By - Abhishek Sir



Recap of Previous Lecture



Topic

Subnetting

in classful N/w





Topics to be Covered



Topic

Subnetting

in classless N/w



ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : https://t.me/abhisheksirCS_PW



[MCQ]

[GATE-2019][2 Mark]



#Q. Consider three machines M, N and P with IP addresses "100 . 10 . 5 . 2", "100 . 10 . 5 . 5" and "100 . 10 . 5 . 6" respectively. The subnet mask is set to "255 . 255 . 255 . 252" for all the three machines. Which one of the following is true?

- ☐ A M, N and P all belong to the same subnet FALSE
- ☒ B Only N and P belong to the same subnet TRUE
- ☐ C Only M and N belong to the same subnet FALSE
- ☐ D M, N, and P belong to three different subnets FALSE

Ans: B

Subnet Mask :

$$255.255.255.252 = 255.255.255.11111100$$

Prefix (30 bit)
2 bit Host ID

$$\underline{M} : 100.10.5.2 = 100.10.5.00000010$$

$$\underline{N} : 100.10.5.5 = 100.10.5.000000101$$

$$\underline{P} : 100.10.5.6 = 100.10.5.000000110$$

[MCQ]

[GATE-IT-2006] [2 Mark]



#Q. A subnetted Class B network has the following broadcast address : 144 . 16 . 95 . 255. Its subnet mask

- ☒ A is necessarily 255 . 255 . 224 . 0
- ☒ B is necessarily 255 . 255 . 240 . 0
- ☒ C is necessarily 255 . 255 . 248 . 0
- ☒ D could be any one of 255 . 255 . 224 . 0, 255 . 255 . 240 . 0, 255 . 255 . 248 . 0

Ans: D

CLASS B :-



Subnet Directed Broadcast Address :

$$144.16.95.255 = 144.16.01011111.11111111$$

16 bit NetID Min^m 3 bit SubnetID Max^m 13 bit HostID

Subnet Mask - (should contains atleast 19 one's)

$$255.255.224.0 = 255.255.11100000.00000000$$

$$255.255.240.0 = 255.255.11110000.00000000$$

$$255.255.\cancel{240}.0 = 255.255.11110000.00000000$$

248 5 bit SubnetID 11 bit HostID



Topic : Size of network



→ Network Size : Maximum possible number of hosts can be in a network

For Host IP address (32-bits)

NetID field (x bits)

→ Subnet ID field (y_1 bits)

HostID field (y_2 bits)

$$\text{Network Size} = [\text{Number of Subnets per network}] * [\text{Sub-network size}]$$

$$= [2^{y_1}] * [2^{y_2} - 2] \text{ hosts per network}$$



Topic : Reduction in networks size



Network Size (before subnetting) $(y = y_1 + y_2)$

$$S_1 = [2^{(y_1 + y_2)} - 2] \text{ hosts per network}$$
$$= (2^y - 2) \text{ hosts per network}$$

Network Size (after subnetting)

$$S_2 = [2^{y_1}] \text{ subnets per network} * [2^{y_2} - 2] \text{ hosts per subnet}$$
$$= [2^{(y_1 + y_2)} - 2^{(y_1 + 1)}] \text{ hosts per network}$$



Topic : Reduction in networks size



→ Subnet ID field (y_1 bits)

Reduction in Network Size (after subnetting)

$$[S_1 - S_2] = [2^{(y_1+1)} - 2] \text{ hosts per network}$$

$$= (2 * 2^{y_1} - 2) \text{ hosts}$$

Classless network (Network Address : 75 . 80 . 160 . 0 / 20), with 4-bit subnetting.

Network Address : 75 . 80 . 160 . 0 / 20

Broadcast Address : 75 . 80 . 175 . 255 / 20
(Network Directed)

10100000.00000000
12 bit Host ID

10101111.11111111
12 bit Host ID

Before Subnetting :-

First Host IP Address : 75 . 80 . 160 . 1 / 20

Last Host IP Address : 75 . 80 . 175 . 254 / 20

Network Size = $(2^{12} - 2)$ hosts

10100000.000000001
10101111.111111110



After 4-bit Subnetting :-

Sub-network Address : 75.80.1010 ^{20 bit NetID} ^{4 bit SubnetID} ^{8 bit HostID} _ _ _ _ . 00000000 / 24

First Host IP Address : 75.80.1010 _ _ _ _ . 00000001 / 24

Last Host IP Address : 75.80.1010 _ _ _ _ . 11111110 / 24

Subnet Broadcast Address : 75.80.1010 _ _ _ _ . 11111111 / 24
(Sub-network Directed)

$$\left[\frac{2^{12} \text{ numbers}}{2^4 \text{ subnets}} = 2^8 \text{ numbers per subnet} \right]$$

$$\text{Reduction} = (2 * 2^4 - 2) = 30 \text{ hosts}$$



$$S_1 = (2^{12} - 2) \text{ hosts} = 4094 \text{ hosts}$$

$$S_2 = 2^4 * (2^8 - 2) \text{ hosts} = (2^{12} - 2^5) \text{ hosts}$$

$$S_1 - S_2 = (2^5 - 2) \text{ hosts}$$

First Sub-network Address : 75 . 80 . 1 0 1 0 0 0 0 0 . 0 0 0 0 0 0 0 0 / 24

: 75 . 80 . 160 . 0 / 24

First Host IP Address : 75 . 80 . 1 0 1 0 0 0 0 0 . 0 0 0 0 0 0 0 1 / 24

: 75 . 80 . 160 . 1 / 24

Last Host IP Address : 75 . 80 . 1 0 1 0 0 0 0 0 . 1 1 1 1 1 1 1 0 / 24

: 75 . 80 . 160 . 254 / 24

First Subnet Broadcast Address : 75 . 80 . 1 0 1 0 0 0 0 0 . 1 1 1 1 1 1 1 1 / 24

(Sub-network Directed) : 75 . 80 . 160 . 255 / 24

Second Sub-network Address : 75 . 80 . 1 0 1 0 0 0 0 1 . 0 0 0 0 0 0 0 0 / 24

: 75 . 80 . 1 6 1 . 0 / 24

First Host IP Address : 75 . 80 . 1 0 1 0 0 0 0 1 . 0 0 0 0 0 0 0 1 / 24

: 75 . 80 . 1 6 1 . 1 / 24

Last Host IP Address : 75 . 80 . 1 0 1 0 0 0 0 1 . 1 1 1 1 1 1 1 0 / 24

: 75 . 80 . 1 6 1 . 254 / 24

Second Subnet Broadcast Address: 75 . 80 . 1 0 1 0 0 0 0 1 . 1 1 1 1 1 1 1 1 / 24

(Sub-network Directed) : 75 . 80 . 1 6 1 . 255 / 24

Last Sub-network Address : 75 . 80 . 1 0 1 0 | | | | . 0 0 0 0 0 0 0 0 / 24

: 75 . 80 . 1 75 . 0 / 24

First Host IP Address : 75 . 80 . 1 0 1 0 | | | | . 0 0 0 0 0 0 0 1 / 24

: 75 . 80 . 1 75 . 1 / 24

Last Host IP Address : 75 . 80 . 1 0 1 0 | | | | . 1 1 1 1 1 1 1 0 / 24

: 75 . 80 . 1 75 . 254 / 24

Last Subnet Broadcast Address : 75 . 80 . 1 0 1 0 | | | | . 1 1 1 1 1 1 1 1 / 24

(Sub-network Directed) : 75 . 80 . 1 75 . 255 / 24

First Sub-network Address : 75 . 80 . 160 . 0 / 24

Second Sub-network Address : 75 . 80 . 161 . 0 / 24

Third Sub-network Address : 75 . 80 . 162 . 0 / 24

Fourth Sub-network Address : 75 . 80 . 163 . 0 / 24

15th Sub-network Address : 75 . 80 . 174 . 0 / 24

Last (16th) Sub-network Address : 75 . 80 . 175 . 0 / 24

Classless network (Network Address : 40 . 50 . 64 . 0 / 19), with 4-bit subnetting.

Network Address : 40 . 50 . 64 . 0 / 19

Broadcast Address : 40 . 50 . 95 . 255 / 19
(Network Directed)

010000000.0000000000
01011111.11111111
13 bit Host ID

Before Subnetting :-

First Host IP Address : 40 . 50 . 64 . 1 / 19

Last Host IP Address : 40 . 50 . 95 . 254 / 19

Network Size = $(2^{13} - 2)$ hosts



After 4-bit Subnetting :-

16 bit Net ID 4 bit subnet ID 3 bit Host ID

Sub-network Address

: 40 . 50 . 0 1 0 _ _ _ _ 0 . 0 0 0 0 0 0 0 0 / 23

First Host IP Address

: 40 . 50 . 0 1 0 _ _ _ _ 0 . 0 0 0 0 0 0 0 1 / 23

Last Host IP Address

: 40 . 50 . 0 1 0 _ _ _ _ 1 . 1 1 1 1 1 1 1 0 / 23

Subnet Broadcast Address
(Sub-network Directed)

: 40 . 50 . 0 1 0 _ _ _ _ 1 . 1 1 1 1 1 1 1 1 / 23

First Sub-network Address : 40 . 50 . 0 1 0 0 0 0 0 0 . 0 0 0 0 0 0 0 0 / 23

: 40 . 50 . 64 . 0 / 23

First Host IP Address : 40 . 50 . 0 1 0 0 0 0 0 0 . 0 0 0 0 0 0 0 1 / 23

: 40 . 50 . 64 . 1 / 23

Last Host IP Address : 40 . 50 . 0 1 0 0 0 0 0 1 . 1 1 1 1 1 1 1 0 / 23

: 40 . 50 . 65 . 254 / 23

First Subnet Broadcast Address : 40 . 50 . 0 1 0 0 0 0 0 1 . 1 1 1 1 1 1 1 1 / 23

(Sub-network Directed) : 40 . 50 . 65 . 255 / 23

40.50.64.0

⇒ Subnet Add.
(1st)

1st host ← 64.1
64.2

⋮
64.255
65.0
65.1
65.2
⋮

last host ← 65.254

40.50.65.255

⇒ Subnet Broadcast Add.
(1st)

⇒ $(2^9 - 2)$ hosts
per
Subnet



NetID SubnetID HostID

Second Sub-network Address : 40.50.010 0001 0.00000000 / 23
: 40.50.66.0 / 23

First Host IP Address : 40.50.010 0001 0.000000001 / 23
: 40.50.66.1 / 23

Last Host IP Address : 40.50.010 0001 1.11111110 / 23
: 40.50.67.254 / 23

Second Subnet Broadcast Address: 40.50.010 0001 1.11111111 / 23
(Sub-network Directed) : 40.50.67.255 / 23



Last Sub-network Address

Net ID Subnet ID Host ID
: 40 . 50 . 0 1 0 0 . 0 0 0 0 0 0 0 0 / 23
: 40 . 50 . 94 . 0 / 23

First Host IP Address

: 40 . 50 . 0 1 0 0 . 0 0 0 0 0 0 0 1 / 23
: 40 . 50 . 94 . 1 / 23

Last Host IP Address

: 40 . 50 . 0 1 0 1 . 1 1 1 1 1 1 1 0 / 23
: 40 . 50 . 95 . 254 / 23

Last Subnet Broadcast Address

(Sub-network Directed)

: 40 . 50 . 0 1 0 1 . 1 1 1 1 1 1 1 1 / 23
: 40 . 50 . 95 . 255 / 23

First Sub-network Address : 40 . 50 . 64 . 0 / 23

Second Sub-network Address : 40 . 50 . 66 . 0 / 23

Third Sub-network Address : 40 . 50 . 68 . 0 / 23

Fourth Sub-network Address : 40 . 50 . 70 . 0 / 23

15th Sub-network Address : 40 . 50 . 92 . 0 / 23

Last (16th) Sub-network Address : 40 . 50 . 94 . 0 / 23



Classless network (Network Address : 90 . 152 . 0 . 0 / 14), with 5-bit subnetting.

(H.W.)

Network Address : 90 . 152 . 0 . 0 / 14

Broadcast Address : 90 .
(Network Directed)

Before Subnetting :-

First Host IP Address : 90 .

Last Host IP Address : 90 .

Network Size =

After 5-bit Subnetting :-

Sub-network Address : 90.100110___.00000.000000000 / 19

First Host IP Address : 90.100110___.00000.000000001 / 19

Last Host IP Address : 90.100110___.11111.111111110 / 19

Broadcast Address : 90.100110___.11111.111111111 / 19
(Sub-network Directed)



2 mins Summary



Topic

Subnetting

(Fixed-Length subnetting)



THANK - YOU

